

Honours Degree of Bachelor of Science in Artificial Intelligence

Batch 21 - Level 2 (Semester 2)

CM 2320 - Mathematical Methods

Tutorial 10

1. Find the Fourier series representation of $f(t) = 1 + t$, $-\pi < t \leq \pi$, period 2π .
-

2. Find the Fourier series representation of the function with period 2π defined by $f(t) = t^2$, $0 < t \leq 2\pi$.
-

3. Find the Fourier series representation of the function

$$f(t) = \begin{cases} 0, & -5 < t < 0, \\ 1, & 0 < t < 5. \end{cases}$$

Given that the period is 10.

4. Find the Fourier series representation of the function

$$f(t) = \begin{cases} -t, & -\pi < t < 0, \\ 0, & 0 < t < \pi. \end{cases}$$

Given that the period is 2π .

5. Find the Fourier series representation of the function

$$f(t) = t^2 + \pi t, \quad -\pi < t < \pi.$$

Given that the period is 2π .

6. Find the Fourier series representation of the function

$$f(t) = \begin{cases} -4, & -\pi < t \leq 0, \\ 4, & 0 < t < \pi. \end{cases}$$

Given that the period is 2π .
